Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 5 Handout

Thursday, September 24

Problem 5.1

A transmitter sends data packets over a computer network. Let T be the travel time of a packet and S be its size. A packet contains 10^2 bytes with probability $5/6$ and 10^4 bytes with probability $1/6$. The travel time T of the data packet depends on its size S and on the congestion in the network at the time of transmission. In particular, the travel time is $10^{-3} \cdot S$ seconds with probability $1/3$ and $10^{-1} \cdot S$ seconds with probability $2/3$.

(a) Determine the PMF of S.

(b) Determine the conditional PMF of T given S for each value of S.

(c) Determine the marginal PMF of T.

(d) Determine the expected travel time for a data packet chosen at random.

(e) We define $U \triangleq S/T$ to be the bandwidth at which the packet was transmitted. Determine the expected bandwidth of the data packets that are transmitted over this network.

Problem 5.2

Let $X \in \mathcal{X} = \{\texttt{rainy}, \texttt{sunny}\}$ be the weather in Massachusetts. Assume X is equally likely to be rainy (`rainy`) or sunny (`sunny`).

(a) Determine the PMF and entropy of X.

Let $Y \in \mathcal{X}$ be the weather in New Hampshire. Since the two states are close, if the weather in Massachusetts is $X = x$, then the weather in New Hampshire is $Y = x$ with probability $1 - \epsilon$, where $\epsilon < 1/2$ is a known small positive constant.

(b) Determine the joint PMF of X and Y.

(c) Determine the distribution of the weather in Massachusetts given that it is a sunny day in New Hampshire.

(d) Determine the conditional entropy of X given Y.